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Method for setting three-dimensional image display system

Abstract

A second controller causes a display panel to sequentially display a plurality of test images, and causes a second camera in an external camera to sequentially capture the plurality of test images to obtain a plurality of second images. The captured second images are transmitted from the external camera to a three-dimensional image display system. The three-dimensional image display system extracts luminance from the received second images and creates luminance plots based on the extracted luminance. A third controller calculates phase values based on the created luminance plots. The third controller stores the calculated phase values with corresponding coordinates on the display panel into a storage as a phase distribution.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
10996472	12/2020	Beckman	N/A	H04N 23/55
2009/0115586	12/2008	Lvovskiy	340/438	G02B 27/01
2015/0177591	12/2014	Sugiyama	359/298	G02B 27/0093
2016/0209647	12/2015	Fürsich	N/A	G02B 27/0075
2016/0291330	12/2015	Kurokawa	N/A	G02B 27/0179
2018/0041699	12/2017	Tohara	N/A	H04N 13/344
2018/0264940	12/2017	Torii	N/A	G08G 1/166
2019/0271840	12/2018	Kishigami	N/A	B60K 35/233
2020/0201037	12/2019	Yamamoto	N/A	B60K 35/00
2021/0107356	12/2020	Watanabe	N/A	B60K 35/81
2021/0152812	12/2020	Ota	N/A	G02B 27/0179
2021/0364810	12/2020	Kim	N/A	G02B 27/0101
2022/0091415	12/2021	Misawa	N/A	A61B 5/163
2023/0066670	12/2022	Yasuda	N/A	G08B 21/06
2024/0001761	12/2023	Dehkordi	N/A	B60K 35/00
2024/0061240	12/2023	Hirata	N/A	B60R 11/02
2024/0126095	12/2023	Shintani	N/A	B60R 1/29
2024/0198908	12/2023	Shintani	N/A	H04N 9/3188
2024/0253465	12/2023	Shintani	N/A	H04N 13/363

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Background/Summary

TECHNICAL FIELD

(1) The present disclosure relates to a method for setting a three-dimensional (3D) image display system.

SUMMARY

(2) In one embodiment of the present disclosure, a method is for setting a three-dimensional image display system. The three-dimensional image display system includes a display panel that displays a parallax image including a right-eye image and a left-eye image, an optical panel that defines a traveling direction of light for each of the right-eye image and the left-eye image, a reflective plate that reflects the right-eye image and the left-eye image with the traveling direction of light being defined, a first camera that captures a first image including an image of an area expected to include a face of a user of the three-dimensional image display system, a first controller that detects a position of at least one of eyes of the user as a first position based on the first image, a second controller that generates the parallax image including the right-eye image and the left-eye image based on a first distribution of a first parameter and the first position, a third controller that calculates a second distribution of the first parameter, and a first communicator that communicates with an external device. The external device includes an external camera including a second camera that captures a second image and a second communicator that communicates directly or indirectly with the first communicator. The method includes moving the second camera to a position to allow the parallax image reflected from the reflective plate to be captured, detecting the position of the second camera as a second position based on the first image, sequentially displaying a plurality of test images on the display panel, and sequentially capturing the plurality of test images reflected from the reflective plate with the second camera to obtain a plurality of second images, calculating the second distribution based on the plurality of second images and the second position, and replacing the first distribution with the calculated second distribution.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a schematic diagram of an example movable body incorporating a 3D image display system.
- (2) FIG. 2 is a schematic diagram of an example 3D image display system.
- (3) FIG. 3 is a schematic diagram describing the relationship between the eyes of a driver, a display panel, and a barrier.
- (4) FIG. 4A is a schematic front view of the display panel.
- (5) FIG. 4B is a diagram of left-eye viewing areas.
- (6) FIG. 4C is a diagram of right-eye viewing areas.
- (7) FIG. 5A is a diagram of example left-eye viewing areas.
- (8) FIG. 5B is a diagram of left-eye viewing areas with deviating viewable subpixels.
- (9) FIG. 6A is a diagram of an example test image.
- (10) FIG. 6B is a diagram of an example test image.
- (11) FIG. 6C is a diagram of an example test image.
- (12) FIG. 6D is a diagram of an example test image.
- (13) FIG. 6E is a diagram of an example test image.
- (14) FIG. 6F is a diagram of an example test image.
- (15) FIG. 6G is a diagram of an example test image.
- (16) FIG. 6H is a diagram of an example test image.

- (17) FIG. 7A is a diagram illustrating the positions of subpixels in a left-eye viewing area.
- (18) FIG. 7B is a graph showing an example plot indicating luminance.
- (19) FIG. 8A is a diagram illustrating the positions of subpixels in a left-eye viewing area.
- (20) FIG. 8B is a graph showing another example plot indicating luminance.
- (21) FIG. 9 is a diagram of an example phase distribution.
- (22) FIG. 10 is a flowchart of a setting process performed by the 3D image display system.
- (23) FIG. 11A is a schematic diagram describing a method for setting distortion correction information.
- (24) FIG. 11B is a schematic diagram describing a method for setting distortion correction information.

DESCRIPTION OF EMBODIMENTS

- (25) The objects, features, and advantages of the present disclosure will become more apparent from the following detailed description and the drawings.
- (26) As the structure that forms the basis of the present disclosure, a known three-dimensional (3D) image display system directs a part of image light from a display panel to reach the right eye of a user and directs another part of the image light to reach the left eye of the user to generate parallax between the two eyes of the user to cause the user to view a 3D image.
- (27) Image light from the display panel travels in the direction defined by, for example, an optical panel, and reaches the right eye or the left eye of the user. The image light from the display panel may also indirectly reach the eyes of the user with, for example, a reflective plate.
- (28) During assembly or installation of the 3D image display system, the optical panel or the reflective plate may have any positional deviation. Although no such positional deviation occurs during assembly or installation or such a positional deviation is negligible in viewing a 3D image, the optical panel or the reflective plate may positionally deviate over time or may deform after use of the 3D image display system. Such positional deviation or deformation may prevent the user from properly viewing a 3D image.
- (29) One or more aspects of the present disclosure are directed to a method for setting a 3D image display system that allows a user to properly view a 3D image.
- (30) One or more embodiments of the present disclosure will now be described in detail with reference to the drawings. The drawings used herein are schematic and are not drawn to scale relative to the actual size of each component.
- (31) A 3D image display system **100** implements a setting method according to one embodiment of the present disclosure. FIG. 1 is a schematic diagram of an example movable body incorporating the 3D image display system. As illustrated in FIG. 1, the 3D image display system **100** may be mounted on a movable body **10**. The 3D image display system **100** may include, for example, a detector **50** and a 3D projector **12**. The 3D image display system **100** communicates with an external camera **101**.
- (32) Examples of the movable body in the present disclosure include a vehicle, a vessel, and an aircraft. Examples of the vehicle include an automobile, an industrial vehicle, a railroad vehicle, a community vehicle, and a fixed-wing aircraft traveling on a runway. Examples of the automobile include a passenger vehicle, a truck, a bus, a motorcycle, and a trolley bus. Examples of the industrial vehicle include an industrial vehicle for agriculture and an industrial vehicle for construction. Examples of the industrial vehicle include a forklift and a golf cart. Examples of the industrial vehicle for agriculture include a tractor, a cultivator, a transplanter, a binder, a combine, and a lawn mower. Examples of the industrial vehicle for construction include a bulldozer, a scraper, a power shovel, a crane vehicle, a dump truck, and a road roller. Examples of the vehicle may include human-powered vehicles. The classification of the vehicle is not limited to the above examples. Examples of the automobile include an industrial vehicle travelling on a road. One type of vehicle may fall within multiple classes. Examples of the vessel include a jet ski, a boat, and a tanker. Examples of the aircraft include a fixed-wing aircraft and a rotary-wing aircraft. In the

example described below, the movable body **10** is a passenger vehicle. The movable body **10** may be any of the above examples rather than a passenger vehicle.

(33) The detector **50** includes a first camera **11A** and a first controller **15**. The first camera **11A** captures a first image including an area expected to include the face of a user **13** of the 3D image display system **100**. In the present embodiment, the user **13** may be, for example, the driver of the movable body **10** that is a passenger vehicle. The area expected to include the face of the driver may be, for example, around an upper portion of the driver's seat. The first camera **11A** may be installed in the movable body **10**. The first camera **11A** may be installed at any position inside or outside the movable body **10**.

(34) The first camera **11A** may be a visible light camera or an infrared camera. The first camera **11A** may function both as a visible light camera and an infrared camera. The first camera **11A** may include, for example, a charge-coupled device (CCD) image sensor or a complementary metal-oxide semiconductor (CMOS) image sensor.

(35) The first image captured with the first camera **11A** is output to the first controller **15**. The first controller **15** detects, based on the first image, the position of at least one of eyes **5** of the user **13** as a first position. The detection result obtained by the detector **50** may be coordinate information indicating the pupil positions of the eyes **5** of the user **13**.

(36) The detector **50** may include, for example, a sensor. The sensor may be, for example, an ultrasonic sensor or an optical sensor. The first camera **11A** may detect the position of the head of the user **13** with the sensor, and may detect the positions of the eyes **5** of the user **13** based on the position of the head. The first camera **11A** may use two or more sensors to detect the positions of the eyes **5** of the user **13** as coordinates defined in a 3D space.

(37) The detector **50** may output coordinate information about the detected pupil positions of the eyes **5** to the 3D projector **12**. The 3D projector **12** may control an image to be projected based on the coordinate information. The detector **50** may output information indicating the pupil positions of the eyes **5** to the 3D projector **12** through wired or wireless communication. The wired communication may include, for example, communication using a controller area network (CAN).

(38) The detector **50** may include the first controller **15** that is an external device. The first camera **11A** may output a captured image to the external first controller **15**. The external first controller **15** may detect the pupil positions of the eyes **5** of the user **13** based on the image output from the first camera **11A**. The external first controller **15** may output the coordinate information about the detected pupil positions of the eyes **5** to the 3D projector **12**. The 3D projector **12** may control an image to be projected based on the coordinate information. The first camera **11A** may output the captured first image to the external first controller **15** through wired or wireless communication. The external first controller **15** may output the coordinate information to the 3D projector **12** through wired or wireless communication. Wired communication may include, for example, communication using a CAN.

(39) The 3D projector **12** may be at any position inside or outside the movable body **10**. For example, the 3D projector **12** may be inside the dashboard in the movable body **10**. The 3D projector **12** emits image light toward a windshield **25**.

(40) The windshield **25** is a reflective plate that reflects image light emitted from the 3D projector **12**. The image light reflected from the windshield **25** reaches an eye box **16**. The eye box **16** is an area defined in a real space in which the eyes **5** of the user **13** are expected to be located based on, for example, the body shape, posture, and changes in the posture of the user **13**. The eye box **16** may have any shape. The eye box **16** may include a two-dimensional (2D) area or a 3D area. The solid arrow in FIG. **1** indicates a path traveled by at least a part of image light emitted from the 3D projector **12** to reach the eye box **16**. The path traveled by image light is also referred to as an optical path. The image light emitted from the 3D projector **12** represents a parallax image including a right-eye image and a left-eye image. With the eyes **5** of the user **13** located in the eye box **16** receiving image light of the parallax image, the user **13** can view a virtual image **14**. The

virtual image **14** is on a path extending forward from the movable body **10** in alignment with the path from the windshield **25** to the eyes **5** (the straight two-dot-dash line in the figure). The 3D projector **12** can function as a head-up display that allows the user **13** to view the virtual image **14**. In FIG. **1**, the direction in which the eyes **5** of the user **13** are aligned corresponds to x-direction. The vertical direction corresponds to y-direction. The imaging range of the first camera **11A** includes the eye box **16**.

(41) As illustrated in FIG. **2**, the 3D projector **12** includes a 3D display device **17** and an optical element **18**. The 3D display device **17** may include a backlight **19**, a display panel **20** including a display surface **20a**, a barrier **21**, and a second controller **24**. The 3D display device **17** may further include a storage **23**. The second controller **24** generates a parallax image to appear on the display panel **20**.

(42) The optical element **18** may include, for example, a first mirror **18a** and a second mirror **18b**. At least either the first mirror **18a** or the second mirror **18b** may have optical power. In the present embodiment, the first mirror **18a** is a concave mirror having optical power. The second mirror **18b** is a plane mirror. The optical element **18** may function as a magnifying optical system that magnifies a parallax image displayed by the 3D display device **17**. The two-dot-dash arrow in FIG. **2** indicates a path traveled by at least a part of image light emitted from the 3D display device **17** to be reflected from the first mirror **18a** and the second mirror **18b** and then exit the 3D projector **12**. The image light emitted from the 3D projector **12** reaches the windshield **25**, is reflected from the windshield **25**, and then reaches the eyes **5** of the user **13**. This allows the user **13** to view the parallax image displayed by the 3D display device **17**.

(43) The optical element **18** and the windshield **25** allow image light emitted from the 3D display device **17** to reach the eyes **5** of the user **13**. The optical system may control the traveling direction of image light to magnify or reduce an image viewable by the user **13**. The optical system may control the traveling direction of image light to deform an image viewable by the user **13** based on a predetermined matrix.

(44) The optical element **18** may have a structure different from the illustrated structure. The mirror may include a concave mirror, a convex mirror, or a plane mirror. The concave mirror or the convex mirror may be at least partially spherical or aspherical. The optical element **18** may be one element or may include three or more elements, in place of two elements. The optical element **18** may include a lens in place of or in addition to a mirror. The lens may be a concave lens or a convex lens. The lens may be at least partially spherical or aspherical.

(45) The backlight **19** is farther from the user **13** than the display panel **20** and the barrier **21** on the optical path of image light. The backlight **19** emits light toward the barrier **21** and the display panel **20**. At least a part of light emitted from the backlight **19** travels along the optical path indicated by the dot-dash line and reaches the eyes **5** of the user **13**. The backlight **19** may include a light emitter such as a light-emitting diode (LED), an organic EL element, or an inorganic EL element. The backlight **19** may have any structure that allows control of the light intensity and the light intensity distribution.

(46) The display panel **20** may be, for example, a liquid-crystal device such as a liquid-crystal display (LCD). In the present embodiment, the display panel **20** includes a transmissive liquid-crystal display panel. The display panel **20** is not limited to this, and may include any of various display panels.

(47) The display panel **20** includes multiple pixels and changes the light transmittance of light from the backlight **19** incident on each pixel to emit image light that then reaches the eyes **5** of the user **13**. The user **13** views an image formed by image light emitted from each pixel in the display panel **20**.

(48) The barrier **21** is an optical panel that defines the traveling direction of incident light. As illustrated in the example in FIG. **2**, with the barrier **21** being nearer the backlight **19** than the display panel **20**, light emitted from the backlight **19** enters the barrier **21** and then enters the

display panel **20**. In this case, the barrier **21** blocks or attenuates a part of light emitted from the backlight **19** and transmits another part of the light to the display panel **20**. The display panel **20** emits incident light traveling in the direction defined by the barrier **21** as image light traveling in the same direction. With the display panel **20** being nearer the backlight **19** than the barrier **21**, light emitted from the backlight **19** enters the display panel **20** and then enters the barrier **21**. In this case, the barrier **21** blocks or attenuates a part of image light from the display panel **20** and transmits another part of the image light to the eyes **5** of the user **13**.

(49) Irrespective of whether the display panel **20** or the barrier **21** is nearer the user **13**, the barrier **21** can control the traveling direction of image light. The barrier **21** allows a part of image light from the display panel **20** to reach one of a left eye **5L** and a right eye **5R** (refer to FIGS. **4A** to **4C**) of the user **13**, and another part of the image light to reach the other of the left eye **5L** and the right eye **5R** of the user **13**. In other words, the barrier **21** directs at least a part of image light toward the left eye **5L** of the user **13** and toward the right eye **5R** of the user **13**. The left eye **5L** is also referred to as a first eye, and the right eye **5R** as a second eye. In the present embodiment, the barrier **21** is located between the backlight **19** and the display panel **20**. In other words, light emitted from the backlight **19** first enters the barrier **21** and then enters the display panel **20**.

(50) The barrier **21** defines the traveling direction of image light to allow each of the left eye **5L** and the right eye **5R** of the user **13** to receive different image light. Each of the left eye **5L** and the right eye **5R** of the user **13** can thus view a different image. This allows the eyes **5** of the user **13** to view a first virtual image **14a** located farther in the negative z-direction than the windshield **25**.

The first virtual image **14a** corresponds to the image appearing on the display surface **20a**. The barrier **21** form a second virtual image **14b** in front of the windshield **25** and nearer the windshield **25** than the first virtual image **14a**. As illustrated in FIG. **2**, the user **13** can view an image with the display panel **20** to be at an apparent position that is the position of the first virtual image **14a** and with the barrier **21** to be at an apparent position that is the position of the second virtual image **14b**.

(51) As illustrated in FIG. **3**, the display panel **20** includes left-eye viewing areas **201L** viewable with the left eye **5L** of the user **13** and right-eye viewing areas **201R** viewable with the right eye **5R** of the user **13** on the display surface **20a**. The display panel **20** displays a parallax image including left-eye images viewable with the left eye **5L** of the user **13** and right-eye images viewable with the right eye **5R** of the user **13**. A parallax image refers to an image projected toward the left eye **5L** and the right eye **5R** of the user **13** to generate parallax between the two eyes of the user **13**. The display panel **20** displays left-eye images on the left-eye viewing areas **201L** and right-eye images on the right-eye viewing areas **201R**. In other words, the display panel **20** displays a parallax image on the left-eye viewing areas **201L** and the right-eye viewing areas **201R**. The left-eye viewing areas **201L** and the right-eye viewing areas **201R** are arranged in u-direction indicating a parallax direction. The left-eye viewing areas **201L** and the right-eye viewing areas **201R** may extend in v-direction orthogonal to the parallax direction, or in a direction inclined with respect to v-direction at a predetermined angle. In other words, the left-eye viewing areas **201L** and the right-eye viewing areas **201R** may be arranged alternately in a predetermined direction including a component in the parallax direction. The pitch between the alternately arranged left-eye viewing areas **201L** and right-eye viewing areas **201R** is also referred to as a parallax image pitch. The left-eye viewing areas **201L** and the right-eye viewing areas **201R** may be spaced from each other or adjacent to each other. The display panel **20** may further include a display area to display a planar image on the display surface **20a**. The planar image generates no parallax between the eyes **5** of the user **13** and is not viewed stereoscopically.

(52) FIG. **4A** is a schematic front view of the display panel. As illustrated in FIG. **4A**, the display panel **20** includes multiple subpixels arranged in a grid in the horizontal and vertical directions. Each subpixel has one of the colors, red (R), green (G), or blue (B). One pixel may be a set of three subpixels with R, G, and B. For example, multiple subpixels included in one pixel are arranged in the horizontal direction. For example, subpixels having the same color are arranged in the vertical

direction. A subpixel group includes multiple subpixels. The subpixel groups are repeatedly arranged in the horizontal direction. The subpixel groups are repeatedly arranged in the vertical direction at positions shifted by one subpixel in the horizontal direction from the corresponding subpixels. In the present embodiment, each subpixel group includes eight subpixels. Each subpixel is assigned with an identification number of, for example, 0 to 7.

(53) FIG. 4B illustrates the left-eye viewing areas **201L**. FIG. 4C illustrates the right-eye viewing areas **201R**. Each left-eye viewing area **201L** includes subpixels numbered 4 in the center of the area and the halves of subpixels numbered 3 and 5. Two subpixels are thus viewable in each left-eye viewing area **201L**. Each right-eye viewing area **201R** includes subpixels numbered 0 in the center of the area and the halves of subpixels numbered 7 and 1. Two subpixels are thus viewable in each right-eye viewing area **201R**. The left-eye viewing areas **201L** and the right-eye viewing areas **201R** with subpixels located in this manner allow the user **13** to view a 3D image properly.

(54) FIG. 5A is a schematic diagram of example left-eye viewing areas. For example, when neither a deviation nor distortion occurs in the relative positions of the display panel **20** and the barrier **21**, or in the relative positions of the display panel **20**, the optical element **18**, and the windshield **25**, as illustrated in FIG. 5A, each left-eye viewing area **201L** includes subpixels numbered 4 in the center of the area and the halves of subpixels numbered 3 and 5. However, positional deviations in the manufacturing processes and positional deviations or distortion over time (hereafter referred to as deviations including manufacturing deviations and distortion) are unavoidable and may cause viewable subpixels in the left-eye viewing areas **201L** to deviate as illustrated in FIG. 5B. Such a positional deviation may prevent the user **13** from properly viewing the 3D image. Thus, for example, the parallax image to appear on the display panel **20** may be corrected based on the deviation. Thus, deviations including manufacturing deviations and distortion are to be measured with a high-precision measurement device. Although the left-eye viewing areas **201L** are described in the above example, the same applies to the right-eye viewing areas **201R**.

(55) In the present embodiment, the 3D image display system **100** communicates with the external camera **101**. The external camera **101** is a device external to the 3D image display system **100**. The 3D display device **17** in the 3D image display system **100** may include, for example, a first communicator **26** that can communicate with an external device and a third controller **22**. The first communicator **26** may include an interface that can communicate with an external device. The first communicator **26** may obtain various sets of information from the external device such as the external camera **101** and output the information to the second controller **24**. The interface that can perform communication in the present disclosure may include, for example, a physical connector and a wireless communication device. The physical connector may include an electric connector for transmission with electric signals, an optical connector for transmission with optical signals, and an electromagnetic connector for transmission with electromagnetic waves. The electric connector may include a connector complying with IEC 60603, a connector complying with the USB standard, or a connector used for an RCA terminal. The electric connector may include a connector used for an S terminal specified by EIAJ CP-121aA or a connector used for a D terminal specified by EIAJ RC-5237. The electric connector may include a connector complying with the HDMI (registered trademark) standard or a connector used for a coaxial cable including a British Naval Connector, also known as, for example, a Baby-series N Connector (BNC). The optical connector may include a connector complying with IEC 61754. The wireless communication device may include a wireless communication device complying with the Bluetooth (registered trademark) standard and a wireless communication device complying with other standards including IEEE 802.1a. The wireless communication device includes at least one antenna.

(56) The external camera **101** includes a second camera **11B**, a second communicator **27**, and a fourth controller **28**. The external camera **101** may be operable by the user **13** and may be, for example, a general-purpose device, such as a digital still camera with communication capabilities, a smartphone, or a tablet device. The second camera **11B** captures a second image. The second

camera **11B** captures images expected to be viewed by the user **13** for allowing the user **13** to check the images. The second camera **11B** may capture a second image after, for example, the user **13** moves the external camera **101** to a position to allow the parallax image reflected from the windshield **25** to be captured. For example, the user **13** on the driver's seat may hold the external camera **101** and capture an image with the second camera **11B**. The second camera **11B** may be located in the eye box **16**. The second camera **11B** may be located at any position, other than in the eye box **16**, to allow an image reflected from the windshield **25** to be captured.

(57) The second communicator **27** communicates directly or indirectly with the first communicator **26**. The second communicator **27** may communicate, for example, in accordance with the same telecommunications standards as the first communicator **26**. The fourth controller **28** controls the second camera **11B** and the second communicator **27**, and transmits the second image captured by the second camera **11B** to the first communicator **26** through the second communicator **27**.

(58) An image reflected from the windshield **25** may be affected by deviations including manufacturing deviations and distortion. The second camera **11B** captures and evaluates the affected image to correct the image to appear on the display panel **20** without directly measuring deviations including manufacturing deviations and distortion. The image appearing on the display panel **20** may be, for example, corrected based on a phase distribution.

(59) A phase value and a phase distribution will now be described. A phase value (first parameter) is obtained with, for example, the following procedure. The display panel **20** sequentially displays multiple test images. The second camera **11B** sequentially captures the test images reflected from the windshield **25**. FIGS. **6A** to **6H** each illustrate an example test image. For example, the test images appear in eight patterns, or patterns 0 to 7. The test image in the pattern 0 appears when the display panel **20** includes the subpixels numbered 0 turned on and the subpixels numbered 1 to 7 turned off. The test image in the pattern 1 appears when the display panel **20** includes the subpixels numbered 1 turned on and the subpixels numbered 0 and 2 to 7 turned off. The test images in patterns 2 to 7 appear in the same or similar manner. The second camera **11B** sequentially captures these test images to obtain eight different second images. The luminance of the second image obtained by capturing the test image in each of the patterns 0 to 7 is extracted and associated with the test image in the corresponding pattern. The user **13** operates the external camera **101** to allow the second camera **11B** to capture test images. The user **13** may operate the external camera **101** every time the test image appearing on the display panel **20** is changed. For example, the display panel **20** displays each test image with its pattern number or outputs a message (e.g., with a display or voice output) indicating the change of the test image to allow the user **13** to notice the timing of the change to operate the external camera **101**.

(60) FIG. **7A** is a diagram illustrating the positions of subpixels in the left-eye viewing area. FIG. **7B** is a graph showing an example plot indicating luminance. When the subpixels in the left-eye viewing area **201L** are located as illustrated in FIG. **7A**, the relationship between the test image in each pattern and the luminance of the second image is plotted as in, for example, FIG. **7B**. The subpixels numbered 4 are located in the center of the left-eye viewing area **201L**. The maximum luminance is thus extracted from the second image obtained by capturing the test image in the pattern 4. The halves of the subpixels numbered 3 and 5 are located in the left-eye viewing area **201L**. The luminance extracted from the second images obtained by capturing the test images in the patterns 3 and 5 is a half of the luminance from the image in the pattern 4. The other subpixels are located outside the left-eye viewing area **201L** and are blocked by the barrier **21**. The luminance of the second images obtained by capturing the test images in the patterns 0 to 2, 6, and 7 is thus zero.

(61) FIG. **8A** is a diagram illustrating the positions of subpixels in the left-eye viewing areas. FIG. **8B** is a graph showing another example plot indicating luminance. When the subpixels in the left-eye viewing area **201L** are located as illustrated in FIG. **8A** due to deviations including manufacturing deviations and distortion, the relationship between the test image in each pattern and the luminance of the second image is plotted as in, for example, FIG. **8B**. The subpixels numbered

4 deviate from the center of the left-eye viewing area **201L**. About 70% of the subpixels numbered 5 are located in the left-eye viewing area **201L**. About 30% of the subpixels numbered 3 are located in the left-eye viewing area **201L**. Although the maximum luminance is extracted from the second image obtained by capturing the test image in the pattern 4, as in the example in FIG. 7B, the luminance extracted from the second image obtained by capturing the test image in the pattern 3 is about 30% of the luminance from the image in the pattern 4. The luminance extracted from the second image obtained by capturing the test image in the pattern 5 is about 70% of the luminance from the image in the pattern 4. The luminance plot in FIG. 8B differs from the luminance plot in FIG. 7B. The second camera **11B** captures multiple test images with the fixed gain and exposure time to obtain the luminance plots.

(62) In the present embodiment, a phase value is a parameter indicating the positional deviation of subpixels in the left-eye viewing area **201L** (or right-eye viewing area **201R**). When the subpixels are located as illustrated in, for example, FIG. 7A, the phase value is 4 without any deviation. FIG. 7B illustrates a luminance plot with a phase value of 4. Deviations including manufacturing deviations and distortion may cause the positions of the subpixels in the left-eye viewing area **201L** (right-eye viewing area **201R**) to deviate. When the subpixels are located as illustrated in, for example, FIG. 8A, the phase value is 4.7. FIG. 8B illustrates a luminance plot with a phase value of 4.7. The relationship between the phase value and the luminance plot may be predetermined based on, for example, experiments.

(63) Multiple second images obtained by capturing the test images are transmitted from the second communicator **27** in the external camera **101** to the first communicator **26** in the 3D image display system **100**. The third controller **22** creates luminance plots with the above procedure based on the received second images. The third controller **22** calculates phase values based on the created luminance plots.

(64) The third controller **22** calculates a phase value for each of all the subpixels on the display panel **20** and obtains the calculation results as a phase distribution. The parallax image to appear on the display panel **20** may be corrected based on the obtained phase distribution. As in the example of the phase distribution illustrated in FIG. 9, the third controller **22** may also divide the display surface **20a** into multiple sections and calculate a representative value (phase value) for each section to obtain a distribution of representative values. In the example in the figure, the display surface **20a** may be divided into 35 sections in 5 rows and 7 columns, and a representative phase value may be calculated for each section. Each representative value may be, for example, the phase value at the center of the section, or may be the phase value at a predetermined position other than the center. The coordinates indicating each representative value may be determined before the second camera **11B** captures test images. For example, the subpixels at positions to be used for luminance extraction for representative value calculation are turned on. The coordinates of the positions of the on-state subpixels are then used as the coordinates indicating the representative value before the second camera **11B** captures test images. After the second camera **11B** captures the test images, the third controller **22** extracts the luminance at the coordinates indicating the representative value and creates luminance plots to calculate representative phase values.

(65) The third controller **22** may store, for example, the calculated phase distribution into the storage **23**. The second controller **24** can correct a parallax image to appear on the display panel **20** by referring to the phase distribution stored in the storage **23**. When obtaining the distribution of the representative values, the second controller **24** can obtain the phase distribution covering all subpixels in the display panel **20** by applying the representative value of each section to the phase values of all subpixels in the section. As described above, deviations including manufacturing deviations and distortion may occur over time during the use of the movable body **10**. Thus, the phase distribution stored in the storage **23** may be updated periodically. For example, the phase distribution may be updated during periodic vehicle inspections at maintenance workshops.

(66) FIG. 10 is a flowchart of a setting process performed by the 3D image display system. Before

the process starts, the user **13** moves the external camera **101** to a position to allow the parallax image reflected from the windshield **25** to be captured. For example, the user **13** on the driver's seat may hold the external camera **101** in front of the eyes **5** and operate the external camera **101**. Once the setting process starts, the second controller **24** first causes the display panel **20** to display a test image in step **S1**. The test image is, for example, in the pattern 0. The user **13** operates the external camera **101** to capture the test image in the pattern 0 with the second camera **11B**. The captured second image is transmitted from the external camera **101** to the 3D image display system **100**. When the second image is received in step **S2**, the determination is performed in step **S3** as to whether the second images in all the patterns are received. When the second images in all the patterns are received, the processing advances to step **S4**. When any second image in one or more patterns is yet to be received, the processing returns to step **S1**. At the time when the test image appears in the pattern 0, some second images are yet to be received. The test image in the subsequent pattern 1 thus appears in step **S1**. The test images appear sequentially in this manner and are sequentially captured by the second camera **11B**. The second images transmitted from the external camera **101** are received by the 3D image display system **100**. After the second images in all the patterns up to pattern 7 are received, the processing advances to step **S4** to extract the luminance from the second images. In step **S5**, luminance plots are created based on the extracted luminance. In step **S6**, the third controller **22** calculates phase values based on the created luminance plots. The relationships between the luminance plots and the phase values may be prestored in, for example, the storage **23**. The third controller **22** may compare the created luminance plots with the luminance plots stored in the storage **23** and calculate phase values by, for example, pattern matching. In step **S7**, the third controller **22** stores the calculated phase values with the corresponding coordinates on the display panel **20** into the storage **23** as a phase distribution (second distribution). The third controller **22** updates a phase distribution (first distribution) stored in the storage **23** to the new phase distribution. The phase distribution (first distribution) stored in the storage **23** before the update is the phase distribution updated in the previous setting process. When no previous setting process is performed, the phase distribution (first distribution) stored in the storage **23** before the update is the phase distribution stored as an initial value.

(67) Although the second image is transmitted to the 3D image display system **100** every time the second camera **11B** captures a test image in the flowchart in FIG. **10**, the external camera **101** may transmit all the second images to the 3D image display system **100** at once after all the test images are captured.

(68) As described above, the second controller **24** can correct a parallax image to appear on the display panel **20** by referring to the phase distribution stored in the storage **23**. The phase distribution is the distribution with the eyes **5** of the user **13** at reference positions (at an optimum viewing distance). The phase distribution varies when the eyes **5** of the user **13** moves from the reference positions. The second controller **24** refers to the phase distribution stored in the storage **23** and increases or decreases the phase values from the referenced phase distribution based on the distance between the positions of the eyes **5** of the user **13** (first position) detected by the detector **50** and the reference positions. The second controller **24** may correct the parallax image to appear on the display panel **20** based on the phase values that have been increased or decreased.

(69) Second images captured by the second camera **11B** correspond to images viewable with the eyes **5** of the user **13**. The position of the second camera **11B** corresponds to the positions of the eyes **5** of the user **13**. The phase values calculated based on the second images obtained by capturing the test images and the phase distribution are obtained when the eyes **5** of the user **13** are at the position of the second camera **11B**. In the 3D image display system **100**, the parallax image to appear on the display panel **20** changes in response to a change in the positions of the eyes **5** of the user **13**. When the positions of eyes **5** of the user **13** change and the phase distribution is unchanged, the parallax image based on the phase distribution may be corrected insufficiently

depending on the positions of eyes **5** of the user **13**. This may prevent the user **13** from viewing the 3D image properly. The external camera **101** is operable by the user **13**. The second camera **11B** may thus not be fixed at the positions of the eyes **5**. Thus, the 3D image display system **100** detects the position of the second camera **11B** (external camera **101**) when the test images are captured and associates the position of the second camera **11B** with the phase distribution. An image of the second camera **11B** can be captured by the first camera **11A**. When the test image appears on the display panel **20** to be captured by the second camera **11B**, the first controller **15** detects the position of the second camera **11B** as a second position based on the first image captured by the first camera **11A**. The third controller **22** may calculate phase values based on the detected second position and obtain a phase distribution.

(70) Another embodiment will now be described. In the above embodiment, the user **13** notices the timing of the change of a test image and operates the external camera **101** to capture test images with the external camera **101**. The user **13** with such an operational load may obtain an inappropriate phase distribution with any erroneous operation. The external camera **101** may detect the timing of the change of the test image. For a test image on the display panel **20** to be changed to the next test image, for example, the 3D image display system **100** may notify the external camera **101** after the test image is changed to the next test image. For example, the first communicator **26** in the 3D image display system **100** transmits information indicating that the test image has been changed. When the second communicator **27** in the external camera **101** receives the information, the fourth controller **28** detects the change of the test image being displayed. The second camera **11B** then captures a test image after the change. The second communicator **27** transmits the second image obtained by capturing the test image with the second camera **11B** to the 3D image display system **100**. Upon receiving the second image from the external camera **101**, the 3D image display system **100** associates the currently appearing test image with the received second image, and then displays the next test image.

(71) For a test image on the display panel **20** to be changed to the next test image, the test image appears with, for example, a symbol or a figure indicating the change. The fourth controller **28** detects the change of the test image by detecting the symbol or the figure in the second image captured by the second camera **11B**. The second communicator **27** transmits the second image obtained by capturing the test image after the change to the 3D image display system **100**. Upon receiving the second image from the external camera **101**, the 3D image display system **100** associates the currently appearing test image with the received second image, and then displays the next test image.

(72) In the above embodiments, the external camera **101** transmits second images captured by the second camera **11B** to the 3D image display system **100**. This structure involves transmission of a large amount of data, which takes time. When detecting the change of the test image being displayed, the fourth controller **28** extracts first information from the second image obtained by capturing the test image after the change. The second communicator **27** then transmits the first information. The first information has a smaller amount of data than the second image. The first information may indicate luminance extracted from the second image. The 3D image display system **100** receives the first information from the external camera **101** in place of second images. The third controller **22** calculates the phase distribution based on the received first information.

(73) Although the setting method in the above embodiments includes updating the phase distribution using the second camera **11B**, a setting method in another embodiment may include, for example, updating distortion correction information. The distortion correction information is used to correct any distorted shape of the virtual image **14** based on the positions of the eyes of the user **13**. The distortion correction information may include information to deform the image to appear on the display panel **20** to reduce the distortion of the shape of the virtual image **14**. For example, the distortion correction information may include information indicating the shift amount of each of multiple feature points in the image appearing on the display panel **20**. The multiple

feature points in the image may be arranged in any other manner. For example, the feature points may be arranged in a grid at any distance from each other. The multiple feature points may correspond to multiple pixels that display images. The correction using the distortion correction information deforms the image appearing on the display panel **20** to, for example, include each of the multiple feature points shifted by a shift amount indicated by the distortion correction information. For example, the correction using the distortion correction information may deform the image to appear on the display panel **20** uniformly or nonuniformly before the image appears on the display panel **20**. The user **13** can view the virtual image **14** with a smaller distortion with light projected from the image deformed based on the distortion correction information.

(74) FIGS. **11A** and **11B** are schematic diagrams each describing a method for setting the distortion correction information. For example, a white plate is placed in advance at a position at which the virtual image **14** is viewable or at a predetermined distance frontward from the windshield **25**. For the user **13** seated on the driver's seat, a first test image appears on the display panel **20**. The user **13** operates the external camera **101** to capture the first test image with the second camera **11B**. As illustrated in, for example, FIG. **11A**, a first test image **40** appears with white circles **41** in the four corners of the image and a monitoring area **42** visually indicating the status of obtaining correction information in the center of the image. The distortion correction information is obtained by changing the capturing position for the second camera **11B** within a predetermined range and capturing an image including feature points at each of predetermined positions within the range. The white circles **41** in the four corners appear to determine that the entire first test image **40** is viewable as the virtual image **14**. In the external camera **101**, the fourth controller **28** detects the second image obtained by capturing the first test image **40** with the second camera **11B** including the white circles **41** in the four corners and transmits the detection result to the 3D image display system **100**. Upon receiving the detection result, the 3D image display system **100** displays a second test image **43** indicating feature points on the display panel **20**. The second test image **43** indicating the feature points may be, for example, a grid image **44**, as illustrated in FIG. **11B**. The 3D image display system **100** starts detecting the position (second position) of the second camera **11B** continuously with the detector **50** after the first test image **40** appears.

(75) Although the user **13** moves the external camera **101** to allow the second camera **11B** to capture the second test image **43** at every predetermined position, the user **13** may not recognize the predetermined positions. The user **13** may move the external camera **101**, for example, upward, downward, leftward, and rightward in any manner. As described above, the 3D image display system **100** continues to detect the position (second position) of the second camera **11B**. Thus, when the second position matches one of the predetermined positions, the second controller **24** causes the display panel **20** to display the second test image **43** indicating the feature points. In the external camera **101**, the fourth controller **28** detects the appearing second test image **43**, and the second camera **11B** then captures the second test image **43**, which is transmitted to the 3D image display system **100**. Upon receiving the second image from the external camera **101**, the 3D image display system **100** associates the received second image with the second position at the time when the second image **43** appears on the display panel **20** to store the second image into the storage **23**.

(76) When the second position matches no predetermined position or is a position stored in the storage **23** in a manner associated with the second image, the display panel **20** displays the first test image **40**. For the second position stored in the storage **23** in a manner associated with the second image, the position may appear in the monitoring area **42** in the center of the first test image **40** to indicate that the image used to obtain distortion correction information is captured. The user **13** views the monitoring area **42** to visually check whether a second image is captured at the predetermined position. The user **13** can, for example, move the external camera **101** intentionally to place the second camera **11B** at a predetermined position at which no second image is captured.

(77) As described above, the 3D image display system **100** switches between displaying the first test image **40** and displaying the second test image **43**, and receives the second images captured at

all the predetermined positions and transmitted repeatedly by the external camera **101**. When the 3D image display system **100** receives the second images at all the predetermined positions, the third controller **22** calculates the distribution (second distribution) of the shift amount (first parameter) based on the coordinates of the predetermined positions and the second images, and stores the distribution into the storage **23**. At this time, the third controller **22** updates a shift distribution (first distribution) stored in the storage **23** to the new shift distribution. The shift distribution (first distribution) stored in the storage **23** before the update is the shift distribution updated in the previous setting process. When no previous setting process has been performed, the shift distribution (first distribution) stored in the storage **23** before the update is the shift distribution stored as an initial value.

(78) The 3D image display system **100** can display a distortion-corrected parallax image on the display panel **20** using the shift distribution stored in the storage **23**. The user **13** can view the virtual image **14** based on the distortion-corrected parallax image. As described above, the 3D image display system **100** performs the setting process for a shift distribution to correct distortion as well as the setting process for a phase distortion. The user can thus view a 3D image more properly.

(79) The structure according to the present disclosure is not limited to the structure described in the above embodiments, but may be changed or varied variously. For example, the functions of the components are reconfigurable unless any contradiction arises. Multiple components may be combined into a single unit, or a single component may be divided into separate units. For example, the first controller **15**, the second controller **24**, and the third controller **22** may be separate controllers. In another example, a single controller may function as at least two of the first controller **15**, the second controller **24**, or the third controller **22**. In other words, a single controller may serve as two or more of these controllers. For example, the detector **50** including the first camera **11A** and the second camera **11B** may include a single controller that functions as the first controller **15** and the third controller **22**.

(80) The figures illustrating the configurations according to the present disclosure are schematic. The figures are not drawn to scale relative to the actual size of each component.

(81) In the present disclosure, first, second, or others are identifiers for distinguishing the components. The identifiers of the components distinguished with first, second, and others in the present disclosure are interchangeable. For example, the first eye can be interchangeable with the second eye. The identifiers are to be interchanged together. The components for which the identifiers are interchanged are also to be distinguished from one another. The identifiers may be eliminated. The components without such identifiers can be distinguished with reference numerals. The identifiers such as first and second in the present disclosure alone should not be used to determine the order of components or to suggest the existence of smaller number identifiers.

(82) In the present disclosure, x-axis, y-axis, and z-axis are used for ease of explanation and may be interchangeable with one another. The orthogonal coordinate system including x-axis, y-axis, and z-axis is used to describe the structures according to the present disclosure. The positional relationship between the components in the present disclosure is not limited to being orthogonal.

(83) The present disclosure may be implemented in the following forms.

(84) In one embodiment of the present disclosure, a method is for setting a three-dimensional image display system. The three-dimensional image display system includes a display panel that displays a parallax image including a right-eye image and a left-eye image, an optical panel that defines a traveling direction of light for each of the right-eye image and the left-eye image, a reflective plate that reflects the right-eye image and the left-eye image with the traveling direction of light being defined, a first camera that captures a first image including an image of an area expected to include a face of a user of the three-dimensional image display system, a first controller that detects a position of at least one of eyes of the user as a first position based on the first image, a second controller that generates the parallax image including the right-eye image and the left-eye

image based on a first distribution of a first parameter and the first position, a third controller that calculates a second distribution of the first parameter, and a first communicator that communicates with an external device. The external device includes an external camera including a second camera that captures a second image and a second communicator that communicates directly or indirectly with the first communicator. The method includes moving the second camera to a position to allow the parallax image reflected from the reflective plate to be captured, detecting the position of the second camera as a second position based on the first image, sequentially displaying a plurality of test images on the display panel, and sequentially capturing the plurality of test images reflected from the reflective plate with the second camera to obtain a plurality of second images, calculating the second distribution based on the plurality of second images and the second position, and replacing the first distribution with the calculated second distribution.

(85) In one embodiment of the present disclosure, the method for setting a 3D image display system allows the user to properly view a 3D image.

REFERENCE SIGNS

(86) **5** eye (**5L**: left eye, **5R**: right eye) **10** movable body **11A** first camera **11B** second camera **12** 3D projector **13** user **14** virtual image (**14a**: first virtual image, **14b**: second virtual image) **15** first controller **16** eye box **17** 3D display device **18** optical element (**18a**: first mirror, **18b**: second mirror) **19** backlight **20** display panel (**20a**: display surface) **201L** left-eye viewing area **201R** right-eye viewing area **21** barrier **22** third controller **23** storage **24** second controller **25** windshield **28** fourth controller **40** first test image **43** second test image **50** detector **100** 3D image display system **101** external camera

Claims

1. A method for setting a three-dimensional image display system including a display panel configured to display a parallax image including a right-eye image and a left-eye image, an optical panel configured to define a traveling direction of light for each of the right-eye image and the left-eye image, a reflective plate configured to reflect the right-eye image and the left-eye image with the traveling direction of light being defined, a first camera configured to capture a first image including an image of an area expected to include a face of a user of the three-dimensional image display system, a first controller configured to detect a position of at least one of eyes of the user as a first position based on the first image, a second controller configured to generate the parallax image including the right-eye image and the left-eye image based on a first distribution of a first parameter and the first position, a third controller configured to calculate a second distribution of the first parameter, and a first communicator configured to communicate with an external device, the external device including an external camera including a second camera configured to capture a second image and a second communicator configured to communicate directly or indirectly with the first communicator, the method comprising: moving the second camera to a position to allow the parallax image reflected from the reflective plate to be captured; detecting the position of the second camera as a second position based on the first image; sequentially displaying a plurality of test images on the display panel, and sequentially capturing the plurality of test images reflected from the reflective plate with the second camera to obtain a plurality of second images; calculating the second distribution based on the plurality of second images and the second position; and replacing the first distribution with the calculated second distribution.
2. The method according to claim 1, wherein the third controller associates, with a position at which the second image is captured, the second position detected when a test image of the plurality of test images being displayed is changed.
3. The method according to claim 1, wherein the external camera further includes a fourth controller configured to detect a change of a test image of the plurality of test images being displayed, and when the fourth controller detects the change of the test image of the plurality of test

- images being displayed, the second communicator transmits a second image obtained by capturing a test image after the change to the first communicator.
4. The method according to claim 1, wherein the external camera further includes a fourth controller configured to detect a change of a test image of the plurality of test images being displayed, when detecting the change of the test image of the plurality of the test images being displayed, the fourth controller extracts first information from a second image obtained by capturing a test image after the change, and the second communicator transmits the first information to the first communicator, and the third controller calculates the second distribution using the first information in place of the second image.
5. The method according to claim 1, wherein the second camera captures the plurality of test images with a fixed gain and a fixed exposure time.
6. The method according to claim 1, wherein the three-dimensional image display system includes a single controller as at least two of the first controller, the second controller, or the third controller.
7. The method according to claim 1, wherein the three-dimensional image display system further includes an optical element configured to perform an optical process on the right-eye image and the left-eye image with the traveling direction of light being defined and to direct the right-eye image and the left-eye image to the reflective plate.
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